

Xiang (Emily) Zuo

220 Darryl Dr., Campbell, CA 95008, USA

+1 (408) 872-2860 | xiangzuo2012@gmail.com | <https://www.linkedin.com/in/emilyzuo/>

SUMMARY

Extensive industrial machine learning analytic and engineering experience. Strong backgrounds of architect design in machine learning/deep learning enterprise products. Skilled in **natural language processing (NLP)**, **large-scale systems**, and **database**. Technical leadership on managing multiple projects and defining technical direction of emerging AI market.

EDUCATION

Stanford Ignite, Stanford University Graduate School of Business (GSB)	Jan. 2024 - Mar. 2024
Ph.D. in Computer Science, University of South Florida, Florida (USF)	2010 - 2016
M.S. in Computer Science, Huazhong University of Science and Technology (HUST), China	2006 - 2009
B.S. in Computer Science, Huazhong University of Science and Technology (HUST), China	2000 - 2004

WORK EXPERIENCE

Senior Technical Engineering Manager October 2023 - Now
Teradata | Santa Clara, CA

Lead Teradata's global AI & ML product team, setting technical direction and driving execution across Cloud and On-Prem platforms for several product lines:

- Enterprise vector store – architecting large-scale semantic search and RAG systems for enterprise customers
- Small LMs– use open-source models to power agent systems for on-prem environments, with fast, secure inference
- Agentic AI – automating data science and BI workflows with skill-based, harnessed agents
- ML functions – scaling parallel in-database analytics across Teradata Vantage

Teradata Santa Clara, CA
Staff Machine Learning Engineer March 2019 - Now

- Lead a team to develop a customer-facing **TeraAnalyticChat** that enables AI-based **conversations (Q&A)** and **SQL query generation** in database system, empowering users to easily use in-database analytic functions, all via natural language (Azure Open AI and Cognitive Search)
- Build a Teradata **ChatGPT-like** model by **fine-tuning** the **OpenLLaMA** with synthetically-generated data using retrieval augmented generation (RAG) and a search engine, combined with public and employee-created data. **Large language model (LLM)** techniques like **qLora**, **Adapter**, **PEFT** are used to enhance tuning and memory efficiency (PyTorch, Huggingface)
- Develop COVID-19 Q&A system by using **BERT**, intent classification, arima and elastic search by Tensorflow
- Lead (researched, designed, and developed) the project of GLMNet and NaiveBayes, ZTest, FTest, Chisquare, ANOVA, KNN, VectorDistance, SMOTE on **MapReduce** framework of Teradata platform (C/C++)
- Coach new hires and serve as a core interviewer for technical hiring

Teradata Santa Clara, CA
Senior Machine Learning Engineer Aug. 2016 - Feb. 2019

- Design and develop a suit of distributed machine-learning algorithms (e.g., XGBoost, Adaboost, Random Forest, TF-IDF, NaiveBayesTextClassifier, and Hashing Encodings) on MapReduce framework of Teradata Vantage (JAVA)
- Provide strong support for marketing, account, and professional service teams on solving all kinds of customers' issues
- Serve as **POC site manager** for enterprise customers including Verizon and AT&T

SKILLS

Machine Learning Tools and Libraries: **PyTorch**, **Tensorflow**, **HuggingFace**, **LangChain**, **Scikit-learn**

Programming Languages: Python, Java, C/C++, R, UNIX Shell

DBMS: Teradata Database, PostgreSQL

Application: MapReduce, GPU CUDA, ggplot

Tools: Vim, Bash, Docker, Kubernetes, Git

HONORS AWARDS & PATENTS

"Teradata 2024 Team Recognition Award", Company-wide awards	2024
"Teradata 2024 Individual Recognition Award", Company-wide awards	2024
"An Open-Source Alternative to ChatGPT", Teradata Hack Day 2nd Prize (2nd among 83)	June 2023
"Hyperparameter Tuner for Vantage Analytics", Teradata Hack Day 1st Prize (Top 1 among 60)	June 2021
"Hyper-Segment IDR", Teradata Patent	Dec 2020
"COVID-19 Q&A system based on BERT ", Finalist in Teradata Hack Day (5 among 50)	May 2020